Environmental Emergencies

By: Matthew, Razi, Anushka, Ethan

Scenario 1:

58 YOM w/confusion and vomiting after collapsing on roof in the heat

Answer

Heat Stroke

Heat Illness: Cramps, Exhaustion, and Stroke

Heat Cramps

Normal or near normal

Signs & Symptoms

Painful spasms in the legs or abdomen after heavy sweating. Alert, with normal thinking.

Field Care

- Move to shade and rest.
- Give an oral electrolyte drink if alert and not nauseated.
- Gently stretch and massage the muscle.

Heat Exhaustion

Usually under 104°F (40°C)

Signs & Symptoms

Heavy sweating, pale and clammy skin, headache, dizziness, nausea, weakness. Alert

Field Care

- Move to a cool area and lie down.
- Loosen clothing and cool with wet cloths or a fan.
- Give oral fluids if alert.
- Transport if vomiting or not improving.

Heat Stroke EMERGENCY

104°F (40°C)+ with brain dysfunction

Signs & Symptoms

Confusion, agitation, seizure, or coma. Hot skin, dry or sweaty. Fast pulse and low BP.

Field Care

- Cool immediately: ice immersion or ice packs to neck, armpits, groin with wet sheets.
- Manage airway, give O2 and IV fluids.
- No fever reducers.
- **Transport urgently.**

Cold Injuries: Hypothermia and Frostbite

Mild Hypothermia
90–95°F (32–35°C)

Signs & Symptoms
Shivering, confusion, clumsiness, pale cold skin. Still alert.

Field Care
- Remove wet clothing & insulate from ground/wind.
- Warm trunk, not limbs.
- Give warm sweet drink only if fully alert.

Moderate Hypothermia
82–90°F (28–32°C)

Signs & Symptoms
Shivering stops. Drowsy, slurred speech, slow pulse & breathing, poor judgement.

Field Care
- Handle gently.
- Remove wet clothing & insulate.
- Apply warm packs to chest, neck, & groin.
- Give O2 & transport.

Severe Hypothermia
EMERGENCY
Below 82°F (28°C)

Signs & Symptoms
Unresponsive, no shivering, rigid muscles, weak or absent pulse.

Field Care
- Handle very gently to avoid arrhythmia.
- Start CPR if no pulse or breathing.
- Transport immediately.
- ***"Not Dead until warm & dead."***

Frostbite
Core temp usually normal

Fingers, toes, nose, ears, or cheeks

Signs & Symptoms
Numb, hard, white or gray-yellow skin. Deep: waxy skin, blisters, blue/black.

Field Care
- Remove wet items.
- Don't rub, use snow, or break blisters.
- Rewarm in 99–102°F water only if it won't refreeze.
- Loose dry dressings; transport.

ALCO Protocols

Patient Care Policy (General) **Modified On: December 1, 2011**

HYPERTHERMIA / HEAT ILLNESS

- **Routine Medical Care**
- Protect patient from environment.
- If the patient is in extremis, begin treatment prior to secondary survey.
- Consider: the environment, patient age, and pre-existing conditions

1. **SIGNS AND SYMPTOMS OF A HEAT EMERGENCY**
   - ➔ Weakness or exhaustion
   - ➔ Dizziness
   - ➔ Headache
   - ➔ Sweating may or may not be present
   - ➔ Fainting or feeling faint
   - ➔ Rapid heart rate
   - ➔ Muscle cramps
   - ➔ Altered mental status (coma, seizures, delirium)
2. **PREEXISTING CONDITIONS THAT CAN CONTRIBUTE TO A HEAT EMERGENCY:**
   - ► **Psychiatric disorder** (both because of the medications taken and perhaps the patient's poor judgement)
   - ► **Heart disease**
   - ► **Diabetes**
   - ► **Alcohol**
   - ► **Fever**
   - ► **Fatigue**
   - ► **Obesity**
   - ► **Dehydration** (either decreased fluid intake or sweating)
   - ► **Medications**
3. **TREATMENT:**
   - 3.1 **If the patient is conscious:**
     - 3.1.1 Remove patient from hot environment
     - 3.1.2 Loosen or remove clothing
     - 3.1.3 Place in supine position with legs elevated
     - 3.1.4 Administer O_2
     - 3.1.5 Fan the patient
     - 3.1.6 Water may be given if patient is alert, has a gag reflex, and is not nauseated
   - 3.2 **If altered mental status is present:** (see above)
     - 3.2.1 Place on left side and monitor airway
     - 3.2.2 Wet the skin and fan aggressively
     - 3.2.3 Apply cold packs to the axillae, groin and neck (if available)
     - 3.2.4 Administer IV fluid challenge (250-500 mL NS)
     - 3.2.5 Transport immediately

HYPOTHERMIA

- **Routine Medical Care**
- Protect the patient from the environment
- If patient is in extremis, begin treatment prior to secondary survey
- Check skin temperature

1. **INTRODUCTION:** Hypothermia is a reduced core temperature where the cold challenge overwhelms heat production and heat retention factors. The rate of onset can be:
   - 1.1 **Acute** (minutes to hours) e.g. immersion in cold water
   - 1.2 **Sub-acute** (hours)
   - 1.3 **Chronic** (often over several days) Homeless, drug users, alcoholics, and compromised individuals are at high risk. Elderly persons and those taking certain medicines are also at risk. Injured and seriously ill individuals can become hypothermic quickly

   ➔ **Note:** a hypothermic critical trauma patient has a very high mortality and morbidity rate!
2. **SIGNS AND SYMPTOMS OF HYPOTHERMIA:**
   - 2.1 Altered mental status including: confusion, mood changes, and speech difficulties. The patient's judgment may be affecting causing him/her to exhibit inappropriate behaviors such as removing clothing
   - 2.2 Decreased motor function, poor coordination
   - 2.3 Diminished sense of cold sensation
   - 2.4 Pupils that respond slowly or sluggishly
3. **TREATMENT:**
   - 3.1 **General:**
     - 3.1.1 Remove the patient from the cold environment and prevent further heat loss
     - 3.1.2 Remove wet clothing, begin rewarming - cover with blankets, turn up the heat in the ambulance
     - 3.1.3 Do not let the patient walk or exert him/herself
     - 3.1.4 Administer O_2 - titrate to 94-99% SpO_2 (warmed and humidified is preferred)
     - 3.1.5 Closely monitor cardiac rhythm
     - 3.1.6 Check blood glucose levels. Administer glucose as needed (see ALOC **page 35** – adult or **page 68** - pediatric)
     - 3.1.7 Transport immediately
   - 3.2 **BLS:**
     - 3.2.1 CPR should be initiated if there is any doubt about pulselessness
     - 3.2.2 Severely hypothermic patients may appear dead. If you find an unresponsive, hypothermic patient, take time (30-45 seconds) to try and find a pulse before beginning CPR. Chest compressions should be avoided if any signs of life are present
     - 3.2.3 If VT or VF is present, defibrillation should be attempted. If one shock is unsuccessful, subsequent shocks should be deferred

HYPOTHERMIA

- 3.3 **ALS:**
  - 3.3.1 Give fluid challenge with heated N.S. if possible
  - 3.3.2 Do not delay urgent procedures (IV lines and intubation) but perform them "gently." The severely cold heart is sensitive to a variety of stimuli, and fatal dysrhythmias can be caused by forceful treatment efforts
  - 3.3.3 Defer ACLS medications until rewarming occurs (> 30° C / 86° F)

Scenario 2:

28 YOF pulled from Lake Meritt

Answer

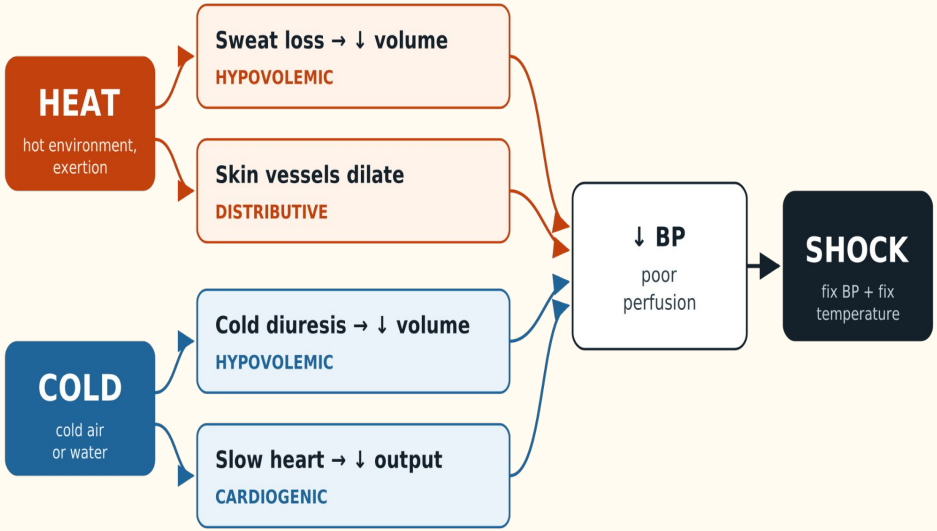
Non-fatal drowning

Thermal/Temperature-Induced Shock

- Not one of the **4 shock types:**

Heat = hypovolemic + distributive, sweat loss lowers fluid and in turn bp, skin vessels dilate to dump heat.
Cold = hypovolemic + cardiogenic, body thinks too much fluid due to vasoconstriction, cold diuresis drops volume, slow/irritable heart drops output and can lead to VF with rough handling, stabilize patient.
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- Standard shock care fails unless temperature is corrected.
- Fast rewarming or lifting upright: → **BP crash.**
- Over-cooling: → **shivering**, makes heat, severely **cold blood can cause cardiac complications.** Cold patient lifted vertically **loses BP.**
- **Cold shock response:** sudden cold water triggers gasp reflex, hyperventilation, HR/BP surge, **underwater gasp → aspiration.**



Thermal/Temperature-Induced Shock

- **Treating Cold:** remove wet clothes, treat heat loss, handle gently, keep horizontal
- **Treating Hot:** remove from heat, aggressive cooling
- ONLY OFFER ORAL FLUIDS IF ALERT AND ORIENTED W/ GAG REFLEX

- **First-aid Scope:** 911, move out of heat/cold, lay them down
- **First-aid Scope Heat** → loosen clothes, dampen skin, fan, ice to neck/armpits/groin
- **First-aid Scope Cold** → dry blankets, warm the trunk, don't rub limbs, CPR if pulseless



Thermal/Temperature-Induced Shock Protocol

Patient Care Policy (General) **Modified On: December 1, 2011**

HYPERTHERMIA / HEAT ILLNESS

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- If the patient is in extremis, begin treatment prior to secondary survey.
- Consider: the environment, patient age, and pre-existing conditions

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   - ▶ **Diabetes**
   - ▶ **Alcohol**
   - ▶ **Fever**
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     - 3.2.3 Apply cold packs to the axillae, groin and neck (if available)
     - 3.2.4 Administer IV fluid challenge (250-500 mL NS)
     - 3.2.5 Transport immediately

AlCo protocol doesn't separate thermal shock. Treat causes with hyper/hypothermia protocol.

Patient Care Policy (General) **Modified On: May 6, 2013**

HYPOTHERMIA

- **Routine Medical Care**
- Protect the patient from the environment
- If patient is in extremis, begin treatment prior to secondary survey
- Check skin temperature

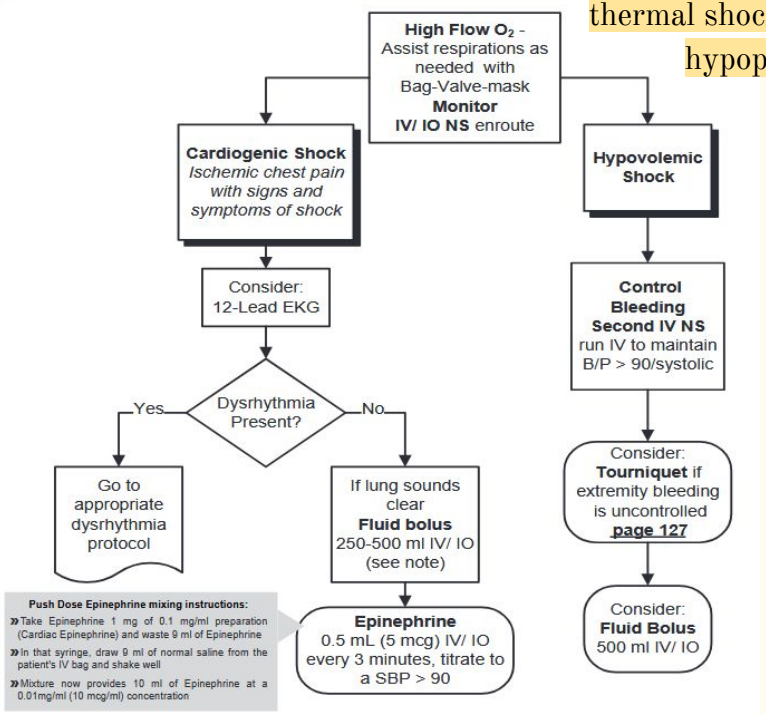
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   - ➔ **Note:** a hypothermic critical trauma patient has a very high mortality and morbidity rate!
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     - 3.2.3 If VT or VF is present, defibrillation should be attempted. If one shock is unsuccessful, subsequent shocks should be deferred

Thermal/Temperature-Induced Shock Protocol

Patient Care Policy (Adult) — Modified On: August 18, 2017

SHOCK: HYPOVOLEMIC/CARDIOGENIC

- **Routine Medical Care**
- **Shock** - 2 or more of the following:
  - → Pulse > 120/minute
  - → BP < 90/systolic
  - → Altered Mental Status
  - → Pale, cool and/or diaphoretic skin signs
- Initiate early transport and treat en route, if appropriate.
- **NOTE:** A fluid bolus of up to 500 ml Normal Saline may be given to an adult patient in cardiogenic shock with clear lung sounds.
- If **anaphylaxis suspected**, see page 36
- If **trauma suspected**, see page 28
- If **sepsis suspected**, see page 56

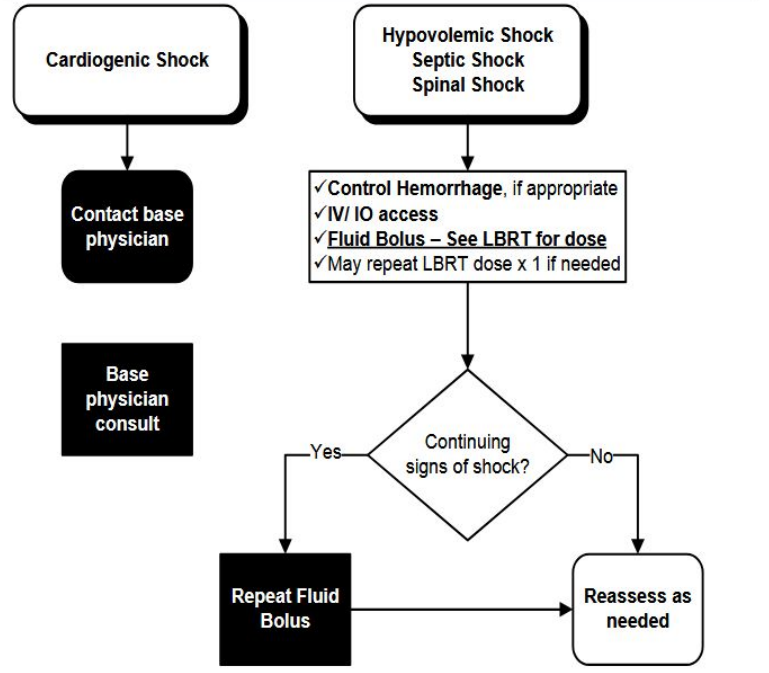


AlCo protocol doesn't separate thermal shock. Treat hypoperfusion with shock protocol.

Patient Care Policy (Pediatric) — Modified On: June 29, 2023

SHOCK AND HYPOTENSION

- **Pediatric Routine Medical Care**
- **IMPORTANT SIGNS OF SHOCK:**
  - → Cool, clammy, mottled skin
  - → Pallor - due to decreased skin perfusion
  - → Altered level of consciousness - due to decreased perfusion to the brain
  - → BP < 70 systolic
- **Initiate early transport and treat en route, if appropriate**
  - → Go to Trauma Patient Care (page 28) if trauma suspected
  - → Go to Allergic Reaction (page 67) if anaphylaxis suspected
- **Use an LBRT to determine pediatric medication dosages** - (Shown underlined on the algorithm)
- **NOTE:** Shock in children may be subtle and hard to recognize. Determining BP may be difficult and readings may be inaccurate



Lightning Strikes & Electric Shocks

- **Scene safety:** power off before touching anyone; lightning can strike the same place twice. Struck patients carry no charge.
- **Lightning = massive DC current in a split second, flashes over skin.**

Lightning injures in 5 ways

direct strike (the bolt hits the person),
side flash (it jumps from a nearby tree or object),
ground current (it spreads through the ground, the most common)
contact (through something the person is touching),
blast (the shock wave throws them).

- **Electric shock = mostly AC, high voltage ≥ 1,000 V.**
- **Fern-like Lichtenberg marks**, temporary limb paralysis, ruptured eardrums, confusion all typical symptoms.
- Entry & exit wounds: small skin burn hides deep damage → kidney injury, compartment syndrome, fractures.





Lightning Strikes & Electric Shocks



- **Tetany:** locks muscles in constant contraction → hand clamps onto the source (“can’t let go”) → longer exposure, possible respiratory arrest, fractures/dislocations.

- Heart may restart, and breathing may not. If so, **ventilate.** Arrest is usually asystole, unshockable.
- **Treat the pulseless/apneic first.** For MCI, Prioritize the worst-looking patients, unless obvious trauma likely to cause death.
- **Late dysrhythmias → transport.**
- **Treat:** Turn off power, AED if pulseless, no cold packs, assume airway involvement, electrical including lightning → burn center.
- **First aid scope:** 911, never touch someone still on a power source, CPR + AED, clean dry cloth on burns, don’t move if thrown/fell.

Lightning Strikes & Electric Shocks Protocol

Patient Care Policy (Adult) **Modified On: January 1, 2026**

BURN PATIENT CARE

Routine Medical Care
- Rescuer safety
- Assume airway/respiratory involvement
- Stop the burning process - *DO NOT USE COLD PACKS*
- Assess for associated trauma

A. BASIC MANAGEMENT

1. **Rule out airway damage**
   - 1.1 Assess for thermal airway injury and smoke inhalation
   - 1.2 High flow oxygen is critical
   - 1.3 Be prepared for intubation
2. **Assess and expose**
   - 2.1 Assess ABCs
   - 2.2 Perform a mini neurological exam - level of consciousness
   - 2.3 Expose and examine the patient for other areas of burn
   - 2.4 Remove jewelry, but do not remove stuck clothing
3. **Start IVs**
   - 3.1 Two large bore IVs (for major burns)
4. **Give IV fluids**
   - 4.1 Any patient regardless of age with suspected 2nd and 3rd degree burns over 20% TBSA, should be given IV with Normal Saline at the below rate. Boluses are discouraged except in cases of shock.
     - ➔ **Adult:** 500 cc/hour (80 drops per minute) or (1-2 drops every second)
     - ➔ **Pediatric:**
       - ≤ 5 years old: 125 cc/hour (20 drops per minute) or (1 drop every 3 seconds)
       - 6-12 years old: 250 cc/hour (40 drops per minute) or (1 drop every 1-2 seconds)
       - ≥ 13 years old: 500 cc/hour (80 drops per minute) or (1-2 drops every second)
5. **Document severity and treat the pain**
   - 5.1 Estimate the severity of the burns using the "Rule of 9s" and "Rule of 1s." The Rule of 1s uses the patient's palm, including fingers, representing approximately 1% of total body surface area, to estimate burn size
   - 5.2 **Treat pain.** Pain management should be considered mandatory for moderate to severe burns. See Pain Management Policies – Adult (page 47) and Pediatric (page 76)
6. **Protect against hypothermia and infection - dress burns**
   - 6.1 Keep patient warm to prevent hypothermia (use sheets or blankets)
   - 6.2 Burns involving less than10% total body surface area (TBSA):
     - ➔ Pour cool running water on the affected area for 20 minutes, if possible
       - Use available tap water (e.g., garden hose) for cooling, sterile water is not required. Do not delay transport to complete the full 20 minutes; cooling does not have to continue in transit.
     - ➔ Apply a dry sterile dressing
   - 6.3 Burns involving greater than or equal to 10% TBSA:
     - ➔ Apply a dry sterile dressing

BURN PATIENT CARE

7. **Elevate burned body parts - 30°**
8. **Address psychological needs**
   - 8.1 Be honest and compassionate
   - 8.2 Consider anxiolytics – **Contact Base Physician for midazolam**
9. **Maintain body temperature and observe for hypothermia**

B. ELECTRICAL BURNS

1. Turn off the power source if patient is still attached
2. See first responder defibrillation protocol if patient is unconscious and pulseless

2. **BURN PATIENT CRITERIA** (from the American Burn Association – Burn Unit Referral Criteria)
   - 2.1 Partial thickness burns greater than 10% total body surface area
   - 2.2 Moderate to severe burns that involve the face, hands, feet, genitalia, perineum, or major joints
   - 2.3 Full thickness burns in any age group
   - 2.4 Electrical burns, including lightning injury
   - 2.5 Chemical burns
   - 2.6 Burn injury in patients with preexisting medical disorders that could complicate management, prolong recovery, or affect mortality
   - 2.7 Inhalation injury

Lightning Strikes & Electric Shocks Protocol

Patient Care Policy (Adult) Modified On: January 1, 2025

CARDIAC ARREST - MEDICAL - ASYSTOLE / PEA

- Routine Medical Care
- Consider and treat other possible causes – See CPR page 12
- If patient presents with signs of obvious death or a valid DNR is presented - See Determination of Death in the Field Procedure

CPR
ECG / AED
BLS Airway (OPA)
BVM ventilation with 10-15 lpm O2
ETCO2 Monitoring

IV / IO NS

Epinephrine 0.1mg/mL
1 mg IV/IO
q 5 minutes, up to 3 doses
(1st dose ASAP – preferably within 5 min from start of CPR)

Consider Advanced Airway

Shockable rhythm?

Yes → Proceed to **Cardiac Arrest – Medical – VF/pVT** protocol

No

2 minutes or 5 cycles of CPR Check rhythm

Organized rhythm and pulse present?

Yes → Proceed to:
Return of Spontaneous Circulation protocol
or
Bradycardia protocol
or
Tachycardia protocol

No → Consider:
Determination of Death Criteria – Medical Arrest Resuscitation
or
Continue CPR Transport

Important Considerations:

- **Do not interrupt CPR to administer medications or perform airway management**
- Use of a mechanical CPR device is required whenever it is available and appropriate
- Consider and treat reversible causes as appropriate:
  - Hypovolemia • Hypoxia • Hydrogen ion (acidosis) • Hypo-/hyperkalemia • Hypothermia
  - Tension pneumothorax • Tamponade, cardiac • Toxins • Thrombosis, pulmonary / coronary
- If renal failure or hyperkalemia suspected, you may consider administering the following:
  - **Calcium Chloride** 1 gm slow IVP (over 2 min.) *Note: flush IV tubing after administering CaCl to avoid precipitation*
  - **Sodium Bicarbonate** 1 mEq/kg IVP *Note: make sure to have a second IV line as other medications may not be compatible*

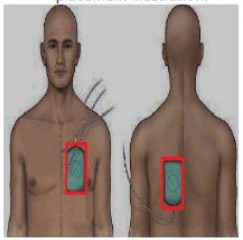
Patient Care Policy (Adult) Modified On: January 1, 2025

CARDIAC ARREST - MEDICAL - VF/PVT

- Routine Medical Care
- Note: Use of a mechanical CPR device is required whenever available and appropriate

***Manual chest compressions**
Place defibrillator pads in the anterior / posterior configuration
Apply mechanical CPR (mCPR) device if available

Anterior/Posterior pad placement illustration:



Shockable rhythm?

No → Proceed to **Cardiac Arrest – Medical – Asystole/PEA** protocol or **Return of Spontaneous Circulation** protocol

Yes

- ****Defibrillate (see note)**
- Resume CPR immediately
- BLS Airway (OPA)
- BVM ventilation at rate of 10-12 with 10-15 lpm O2
- ETCO2 Monitoring
- IV/IO NS
- 2 minutes or 5 cycles of CPR
- Check Rhythm

Shockable rhythm?

No → Proceed to **Cardiac Arrest – Medical – Asystole/PEA** protocol or **Return of Spontaneous Circulation** protocol

Yes

- **CPR** while defibrillator charging
- ****Defibrillate (see note)**
- **Resume CPR**
- *****Epinephrine 0.1mg/mL** 1 mg IV/IO
- 2 minutes or 5 cycles of CPR
- Check Rhythm

Shockable Rhythm?

No → Proceed to **Cardiac Arrest – Medical – Asystole/PEA** protocol or **Return of Spontaneous Circulation** protocol

Yes

- **CPR** while defibrillator charging
- ****Defibrillate (see note)**
- **Resume CPR**
- **Amiodarone** 300 mg IV/IO
- Consider Advanced Airway

- Initiate transport to the closest STEMI center
- Proceed to **Cardiac Arrest – Medical – VF/pVT** protocol

VF/Pulseless VT notes:

***chest compressions:**
CPR/mCPR must be minimally interrupted (<10 secs) and should not be paused for airway placement. mCPR does not need to be paused for defibrillation(s). Manual chest compressors must be rotated at every rhythm check.

****Defibrillation:**
Refer to manufacturer's documentation for energy dose recommendations

*****Epinephrine**
May be repeated q 5mins to a max of 3 doses

Drowning & Secondary Drowning

- **Drowning → hypoxia**: respiratory impairment from submersion.

Breath-hold → laryngospasm/aspiration → hypoxia → bradycardia/LOC →arrest

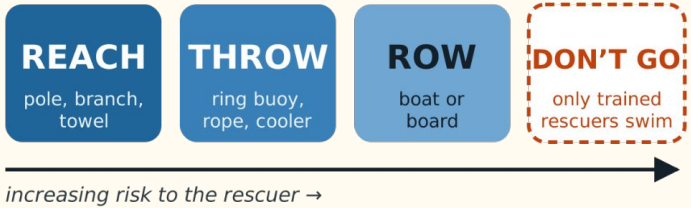
- **In cold water:** gasp reflex & hypothermia, **mammalian diving reflex and hypothermia** slows HR/slows metabolism/shunts blood to brain/heart, protecting the brain.
- **Diving:** Body weight drives down onto the top of the head, compressing the cervical spine. Trauma is likely.
- **SMR only for diving or trauma. In other cases, <1% of cases have c-spine injury.** Ask why/how they went under.

- **Secondary drowning:** delayed symptoms.
- **Aspirated water washes out surfactant** → alveoli collapse and fill with fluid over hours. CPAP can help with this. Not BLS scope.
- **Any symptom after submersion** → transport, even if they feel fine.

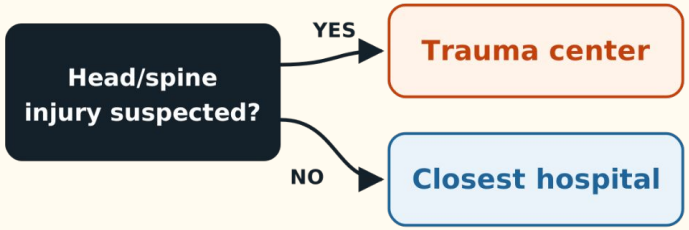
Drowning & Secondary Drowning

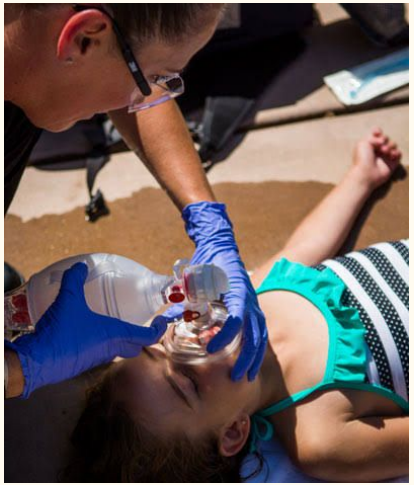
- **Treating:** suction mouth, ventilate through foam, O2 94%+, BVM if inadequate, remove wet clothes, reassess lungs often. Arrest with head/spine injury → trauma center and without → closest hospital
- **First-aid scope:** 911, reach or throw, do NOT go, CPR with rescue breaths, AED, recovery position if breathing, remove wet clothes, any cough after submersion needs medical care.

Bystander water rescue: safest first



Submersion arrest: destination





EMT PRIORITIES

1 Suction mouth → 2 O_2 / BVM, ventilate through foam → 3 Remove wet clothes → 4 Reassess lungs often → 5 Transport every symptomatic patient

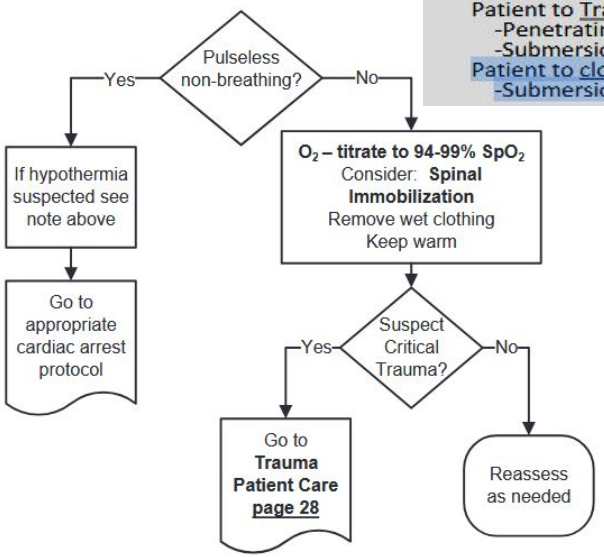
Drowning & Secondary Drowning Protocol

Patient Care Policy (Adult) — Modified On: December 1, 2011

SUBMERSION

- **Routine Medical Care**
- Consider spinal precautions prior to extrication if possibility of neck trauma
- Rapid extrication from water
- **If hypothermia** suspected and the patient is in Ventricular Fibrillation, rapid transport to the closest receiving hospital is essential for rewarming. Patients who are hypothermic rarely respond to treatment. (see Hypothermia **page 19**)
- **Consider CPAP** - see CPAP procedure (**page 124**) for indications

Under Cardiac Trauma:



7. Destination choice:
   Patient to Trauma Center:
   -Penetrating or Blunt trauma with significant hypovolemia from exsanguination
   -Submersion or Mechanical Asphyxiation (drowning, hanging, strangulation) **with** suspected head or spinal injury
   Patient to closest hospital:
   -Submersion or Mechanical Asphyxiation (drowning, hanging, strangulation) **without** suspected head or spinal injury

Procedures — Modified On: December 1, 2011

CONTINUOUS POSITIVE AIRWAY PRESSURE - CPAP

1. **PURPOSE:** To improve ventilation and oxygenation, and avoid intubation. *CPAP is required for all ALS providers.*
2. **INDICATIONS:** Patients age 8 or older in severe respiratory distress and:
   - ► **CHF with pulmonary edema**
   - ► **Near-drowning**
   - ► **Other causes of severe respiratory distress**
3. **CONTRAINDICATIONS** - Bag-valve-mask ventilation or endotracheal intubation should be considered for any patient who exhibits one or more of the following contraindications
   - 3.1 **Absolute Contraindications** (DO NOT USE):
     - ► Age < 8
     - ► Respiratory or cardiac arrest
     - ► Agonal respirations
     - ► Severely depressed level of consciousness
     - ► Systolic blood pressure < 90
     - ► Signs and symptoms of pneumothorax
     - ► Inability to maintain airway patency
     - ► Major trauma, especially head injury with increased ICP or significant chest trauma
     - ► Facial anomalies or trauma (e.g., burns, fractures)
     - ► Vomiting
   - 3.2 **Relative Contraindications** (USE CAUTIOUSLY):
     - ► History of Pulmonary Fibrosis
     - ► Decreased LOC
     - ► Claustrophobia or unable to tolerate mask (after first 1-2 minutes trial)
4. **COMPLICATIONS:**
   - ► Hypotension
   - ► Pneumothorax
   - ► Corneal Drying
5. **GOALS OF CPAP:**
   - ► Elimination of dyspnea
   - ► Decreased respiratory rate
   - ► Decreased heart rate
   - ► Increased SpO_2
   - ► Stabilized blood pressure

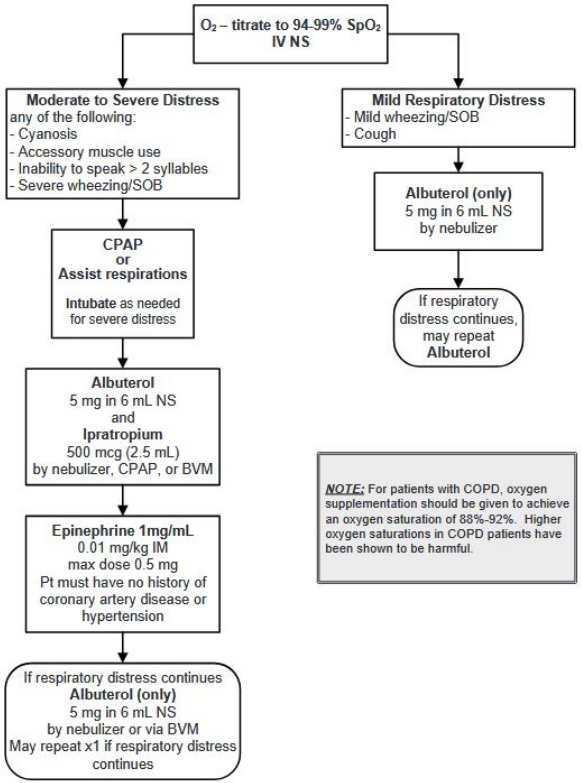
Bag-valve-mask ventilation or endotracheal intubation should be considered if the patient fails to show improvement based on the above goals.

Drowning & Secondary Drowning Protocol

Patient Care Policy (Adult) — Modified On: May 27, 2021

RESPIRATORY DISTRESS

- Routine Medical Care
  - → Asthma
  - → COPD
  - → Bronchospasm
  - → Pulmonary edema (see page 49)
- Limit physical exertion, reduce patient anxiety



Patient Care Policy (General) — Modified On: May 6, 2013

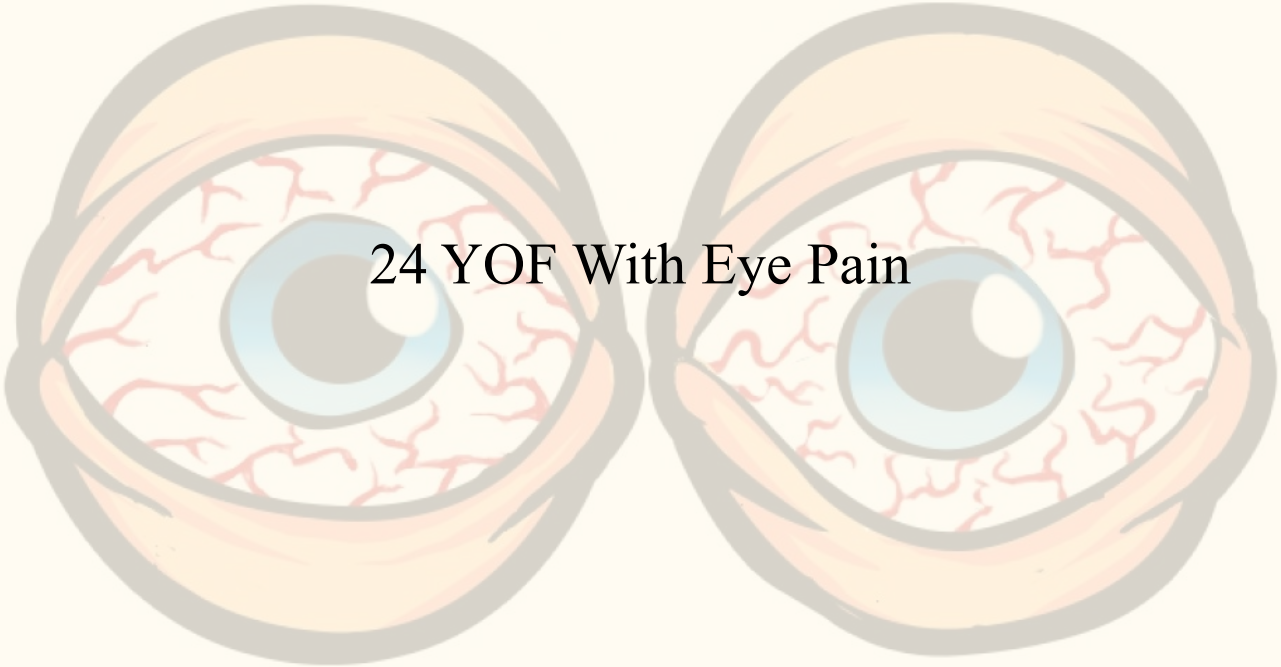
HYPOTHERMIA

- **Routine Medical Care**
- Protect the patient from the environment
- If patient is in extremis, begin treatment prior to secondary survey
- Check skin temperature

1. **INTRODUCTION:** Hypothermia is a reduced core temperature where the cold challenge overwhelms heat production and heat retention factors. The rate of onset can be:
   - 1.1 **Acute** (minutes to hours) e.g. immersion in cold water
   - 1.2 **Sub-acute** (hours)
   - 1.3 **Chronic** (often over several days) Homeless, drug users, alcoholics, and compromised individuals are at high risk. Elderly persons and those taking certain medicines are also at risk. Injured and seriously ill individuals can become hypothermic quickly

   → **Note:** a hypothermic critical trauma patient has a very high mortality and morbidity rate!
2. **SIGNS AND SYMPTOMS OF HYPOTHERMIA:**
   - 2.1 Altered mental status including: confusion, mood changes, and speech difficulties. The patient's judgment may be affecting causing him/her to exhibit inappropriate behaviors such as removing clothing
   - 2.2 Decreased motor function, poor coordination
   - 2.3 Diminished sense of cold sensation
   - 2.4 Pupils that respond slowly or sluggishly
3. **TREATMENT:**
   - 3.1 **General:**
     - 3.1.1 Remove the patient from the cold environment and prevent further heat loss
     - 3.1.2 Remove wet clothing, begin rewarming - cover with blankets, turn up the heat in the ambulance
     - 3.1.3 Do not let the patient walk or exert him/herself
     - 3.1.4 Administer O_2 - titrate to 94-99% SpO_2 (warmed and humidified is preferred)
     - 3.1.5 Closely monitor cardiac rhythm
     - 3.1.6 Check blood glucose levels. Administer glucose as needed (see ALOC page 35 – adult or page 68 - pediatric)
     - 3.1.7 Transport immediately
   - 3.2 **BLS:**
     - 3.2.1 CPR should be initiated if there is any doubt about pulselessness
     - 3.2.2 Severely hypothermic patients may appear dead. If you find an unresponsive, hypothermic patient, take time (30-45 seconds) to try and find a pulse before beginning CPR. Chest compressions should be avoided if any signs of life are present
     - 3.2.3 If VT or VF is present, defibrillation should be attempted. If one shock is unsuccessful, subsequent shocks should be deferred

Scenario 3:



24 YOF With Eye Pain

Answer

Chemical Burn

Burns

Types of Burns

Thermal: Fire, hot liquids, steam

Chemical: Contact with acids, bases, or other chemicals

Electrical: Contact with electricity or lightning

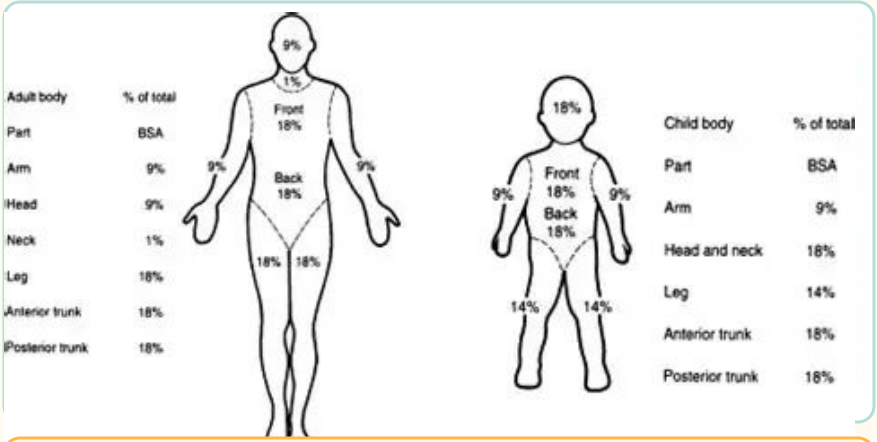
Radiation: Sunburn or other radiation exposure

Assessing Burns

Superficial (1st degree): Red, painful skin; no blisters

Partial-thickness (2nd degree): Blisters, swelling, very painful

Full-thickness (3rd degree): Deep skin damage; may look white, brown, or charred.



Note: *Only count partial-thickness and full-thickness burns when estimating TBSA; don't include simple redness (superficial burns).*

What to do?

Move the person away from whatever caused the burn, cool the burn with cool running water (not ice), and cover it with a clean, dry cloth. Check breathing, estimate rule of nines, watch for signs of shock, transport

ALCO Protocol

BURN PATIENT CRITERIA

1. **INTRODUCTION** -The intent of this policy is to transport patients with critical burns, who have a manageable airway, directly to a facility that is staffed and equipped to care for the medical needs of the patient, bypassing other receiving facilities. Minor to moderate burn patients will be transported to the closest, most appropriate receiving hospital.
2. **BURN PATIENT CRITERIA** (from the American Burn Association – Burn Unit Referral Criteria)
   - 2.1 Partial thickness burns greater than 10% total body surface area
   - 2.2 Moderate to severe burns that involve the face, hands, feet, genitalia, perineum, or major joints
   - 2.3 Full thickness burns in any age group
   - 2.4 Electrical burns, including lightning injury
   - 2.5 Chemical burns
   - 2.6 Burn injury in patients with preexisting medical disorders that could complicate management, prolong recovery, or affect mortality
   - 2.7 Inhalation injury
3. **DESTINATION**
   - 3.1 **Adult and Pediatric patients** who meet burn patient criteria 2.1-2.6 may be transported directly to an out-of-county burn center (see #5 below).
   - 3.2 Exceptions:
     - 3.2.1 **Potentially unmanageable airway** - (e.g. - soot in the mouth and/or nose, inhalation injury, etc.) transport to the closest trauma center.
     - 3.2.2 **Unmanageable Airway** - The patient requires intubation, and the paramedic is unable to intubate, and an adequate airway cannot be maintained with B.V.M. device, transport to closest basic E.D.
     - 3.2.3 **Patient meets Critical Trauma Patient Criteria** – "Mental Status & Vital Signs" - transport to the closest most appropriate designated trauma center
4. **OUT-OF-COUNTY TRANSPORT**
   - 4.1 Transporting medic ***must*** first contact out-of-county hospital to confirm bed availability. This can be done through the appropriate dispatch center or via phone or radio from the field
   - 4.2 Contact the Base Physician if medical consultation is needed
   - 4.3 Consider EMS Aircraft transport for land transport times greater than 45 minutes
   - 4.4 Give a brief report to the receiving facility including ETA

Out-of-County Burn Centers:

FACILITY	ADULT	PEDIATRIC	TRAUMA	HELIPAD	LOCATION	PHONE #
Santa Clara Valley Medical Center	YES	YES	YES	YES	751 S. Bascom Ave., San Jose	(408) 885-6666
UC Davis Medical Center	YES	YES	YES	YES	2315 Stockton Blvd., Sacramento	(916) 734-3636
St. Francis Memorial Hospital	YES	NO	NO	NO	900 Hyde Street, San Francisco	(415) 353-6255

BURN PATIENT CARE

7. **Elevate burned body parts - 30°**
8. **Address psychological needs**
   - 8.1 Be honest and compassionate
   - 8.2 Consider anxiolytics – **Contact Base Physician for midazolam**
9. **Maintain body temperature and observe for hypothermia**

B. ELECTRICAL BURNS

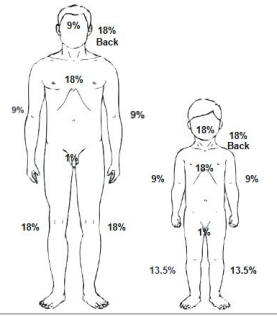
1. Turn off the power source if patient is still attached
2. See first responder defibrillation protocol if patient is unconscious and pulseless

C. TAR BURNS

1. Do not attempt to remove the tar
2. Cool with water
3. **Maintain body temperature and observe for hypothermia**

D. CHEMICAL BURNS

1. Remove clothing
2. **Liquid chemicals:**
   - →Flush **immediately** with copious amounts of tepid water for 15 - 20 minutes
3. **Dry chemicals:**
   - →Brush off as much as possible, then flush with copious amount of tepid water for 10 - 15 minutes
4. Identify chemical
5. Assess for associated respiratory burns

Rule of 9s	Rule of 1s
	Patient's palm and fingers are 1% TBSA 

Decompression Sickness

Happens when a diver comes up to the surface too quickly, causing gas bubbles to form in the body.

Signs & Symptoms

- Joint and muscle pain
- Dizziness, headache, or extreme tiredness
- Numbness, tingling, or weakness
- Trouble breathing or confusion

What EMTs Should Do

- Give high flow oxygen, call for ALS, and Code 3 transport.
- Monitor breathing and vital signs and mental status



Hyperbaric Oxygen Treatment

The patient may need treatment in a specialized chamber to help treat gas bubbles in the body.

Altitude Related Emergencies

HACE — High-Altitude Cerebral Edema

Brain swelling caused by high altitude

Signs & Symptoms:

- Severe headache & confusion
- Poor coordination & difficulty walking
- Unusual behavior & decreased consciousness

HAPE — High-Altitude Pulmonary Edema

Fluid accumulation in lungs at high altitude

Signs & Symptoms:

- Shortness of breath & difficulty breathing at rest
- Cough, chest tightness & weakness
- Sometimes pink or frothy sputum

What EMTs Should Do

- **Give oxygen**
- Monitor breathing, vital signs, and mental status
- Request ALS and assist breathing if needed
- **Arrange evacuation to a lower altitude**

Key Point

HACE affects the **BRAIN**

HAPE affects the **LUNGS**

Crowd Syndrome

Mass gatherings create their own emergencies: heat, intoxication, trauma, and crush injuries.

Crowd Crush

People packed too tight to expand their chest.

Signs & Symptoms:

- Trouble breathing, cyanosis
- Petechiae on face and neck
- Unresponsive after a crowd surge

Mass Psychogenic Illness

One person gets sick, then others nearby report the same symptoms.

Signs & Symptoms:

- Fainting, nausea, dizziness, headache
- Spreads person to person

What EMTs Should Do

- Work from the edges
    - Never enter the dense center
- Stay with your partner and know your exits
- Move patients out of the crowd before assessing
- Assess every patient. Rule out heat, CO, chemicals
- Call for more resources early

Key Point

Crowd crush deaths are from **COMPRESSIVE ASPHYXIA** not trampling

Protocols: Crowd Syndrome

BMRC First Aid Protocol

1. Obtain OPQRST and SAMPLE
2. **Make sure you have enough space to tend to pt**
3. Call UCPD or 911
4. Be ready to give a report to a higher-level-of-care provider

Pit Crew Leader:

- Supervises and assigns crowd control

Bites and Stings

Anaphylaxis

Signs & Symptoms:

- Hives, flushing
- Throat or tongue swelling
- Wheezing, low BP

What EMTs Do:

- EpiPen to lateral thigh
- Adult: 0.3 mg IM
- Child: 0.15 mg IM
- Oxygen
- **Always transport**

Snakes

Venomous:

- Triangular head, slit pupils
- Rattle, fang marks
- Spreading pain, swelling

Non-venomous:

- Rounded head, round pupils
- Treat as a minor wound

What EMTs Do:

- Keep still, remove rings
- Cool, wet cloth
- **Do not elevate**

Note: Rattlesnake is California's main venomous snake. Rattle on the tail.

Spiders (not in ALCO)

Venomous:

- Severe pain, cramping
- Rigid abdomen, sweating

Non-venomous:

- Most spider bites
- Mild redness, itching

What EMTs Do:

- Clean the site
- Cold pack
- Transport if systemic

Note: Black widow is California's main venomous spider. Shiny black, red hourglass on its belly.

Ticks (not in ALCO)

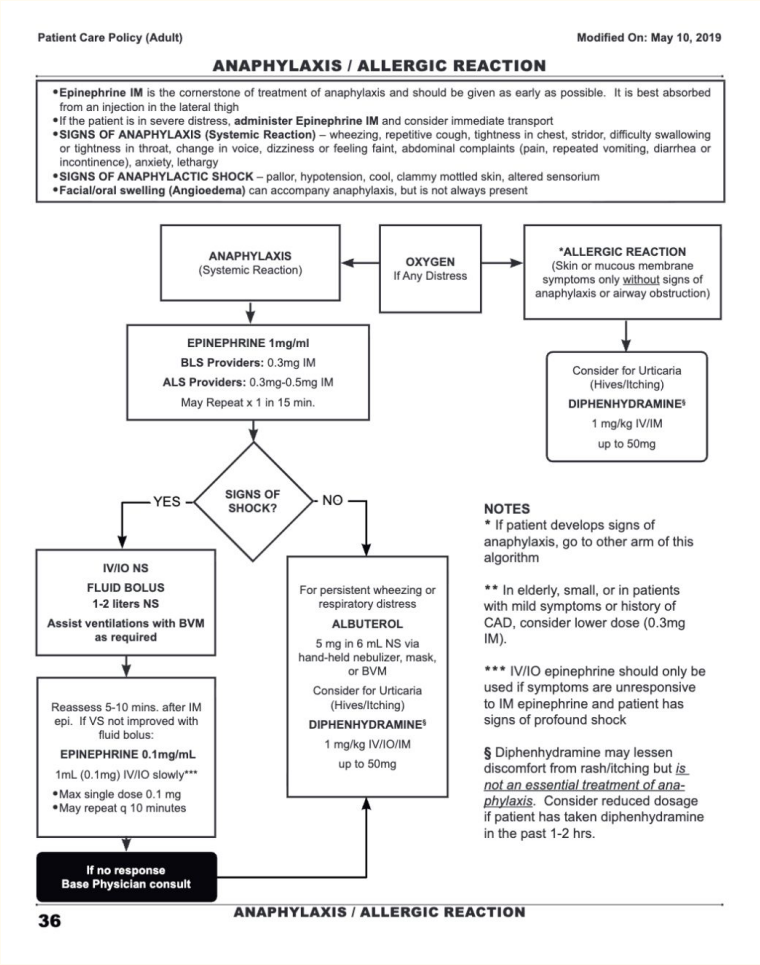
Risk: Lyme disease

Removal:

- Tweezers close to skin
- Steady, straight pull
- No twisting, burning, or Vaseline

Watch for:

- Bullseye rash and flu-like symptoms later



Patient Care Policy (Adult) Modified On: May 10, 2019

ANAPHYLAXIS / ALLERGIC REACTION

- **Epinephrine IM** is the cornerstone of treatment of anaphylaxis and should be given as early as possible. It is best absorbed from an injection in the lateral thigh
- If the patient is in severe distress, **administer Epinephrine IM** and consider immediate transport
- **SIGNS OF ANAPHYLAXIS (Systemic Reaction)** – wheezing, repetitive cough, tightness in chest, stridor, difficulty swallowing or tightness in throat, change in voice, dizziness or feeling faint, abdominal complaints (pain, repeated vomiting, diarrhea or incontinence), anxiety, lethargy
- **SIGNS OF ANAPHYLACTIC SHOCK** – pallor, hypotension, cool, clammy mottled skin, altered sensorium
- **Facial/oral swelling (Angioedema)** can accompany anaphylaxis, but is not always present

ANAPHYLAXIS (Systemic Reaction)

OXYGEN If Any Distress

***ALLERGIC REACTION** (Skin or mucous membrane symptoms only without signs of anaphylaxis or airway obstruction)

EPINEPHRINE 1mg/ml
BLS Providers: 0.3mg IM
ALS Providers: 0.3mg-0.5mg IM
May Repeat x 1 in 15 min.

Consider for Urticaria (Hives/Itching)
DIPHENHYDRAMINE§
1 mg/kg IV/IM
up to 50mg

SIGNS OF SHOCK? YES NO

IV/IO NS
FLUID BOLUS
1-2 liters NS
Assist ventilations with BVM as required

Reassess 5-10 mins. after IM epi. If VS not improved with fluid bolus:
EPINEPHRINE 0.1mg/mL
1mL (0.1mg) IV/IO slowly***
•Max single dose 0.1 mg
•May repeat q 10 minutes

If no response Base Physician consult

For persistent wheezing or respiratory distress
ALBUTEROL
5 mg in 6 mL NS via hand-held nebulizer, mask, or BVM
Consider for Urticaria (Hives/Itching)
DIPHENHYDRAMINE§
1 mg/kg IV/IO/IM
up to 50mg

NOTES
* If patient develops signs of anaphylaxis, go to other arm of this algorithm

** In elderly, small, or in patients with mild symptoms or history of CAD, consider lower dose (0.3mg IM).

*** IV/IO epinephrine should only be used if symptoms are unresponsive to IM epinephrine and patient has signs of profound shock

§ Diphenhydramine may lessen discomfort from rash/itching but *is not an essential treatment of anaphylaxis*. Consider reduced dosage if patient has taken diphenhydramine in the past 1-2 hrs.

36 ANAPHYLAXIS / ALLERGIC REACTION

6. SNAKE BITE:

6.1. Gently wash the area with cool, wet cloth

6.2. Avoid constricting bands

6.3. Do not elevate

Never for snake bites: tourniquet, ice, cutting, or sucking out venom

Scrape bee stingers off with a card. Squeezing injects more venom.

BMRC First Aid Protocol

1. Obtain OPQRST and SAMPLE
2. Ask about **allergy history**
3. **Assist** pt with their own prescribed EpiPen
4. Place pt in a position of comfort
5. Closely monitor vitals every 5 minutes
6. Call UCPD or 911
7. Be ready to give a report to a higher-level-of-care provider

Scene Safety

Priority: **1. You** → **2. Your Partner** → **3. Bystanders** → **4. Patient**

Lightning

- Get indoors
- Wait 30 min after last thunder

Water

- Reach, throw, row, go
- Don't swim unless trained

Electrical

- Assume downed lines are live
- Wait for the utility

Hazmat

- Stay upwind, uphill, upstream

Heat & Cold

- Affects rescuers too
- Hydrate, layer up, rotate

What EMTs Should Do

- Size up the scene before patient contact
- Wear PPE every time
- Call for resources you can't provide (fire, utility, hazmat, police)

Key Point

An injured rescuer becomes **ANOTHER PATIENT**

Protocols: Scene Safety

Scene safety	Make sure the environment is safe for rescuers and victim

BMRC First Aid Protocol

1. **Do not enter an unsafe scene. Wear PPE before patient contact**
2. Call necessary resources

BSI Scene Safe?

Kahoot